IGCSE 0607 Practice Questions — E1.15 — Combined Questions & Markscheme

Money

Total Questions: 12 — **Total Marks:** 70 Marks

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  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(9 Qs)*

1 John has \$400. He buys a new camera that costs 255 Euros (). The exchange rate is \$1 = 0.85.
Work out how much money John has left in dollars.

\$ ..... **[3]**

2 The exchange rate between pounds (£) and dollars ($) is £1 = $1.50.
Anna changes £300 into dollars.
She spends $210 on a tablet.
She changes the remaining dollars back into pounds at the same exchange rate.

Work out how many pounds she receives.

£ ..... **[3]**

3 Amira goes on holiday to Japan.
She changes \$850 into Japanese Yen (JPY). The exchange rate is \$1 = 148 JPY.

(a) Calculate how many JPY she receives.

..... JPY **[2]**

(b) In Japan, she spends 45 000 JPY on a camera and 31 500 JPY on food.
At the end of the holiday, she changes her remaining JPY back to dollars (\$). The new exchange rate is \$1 = 152 JPY.
Calculate how many dollars she receives.
Give your answer correct to the nearest cent.

\$ ..... **[3]**

4 A car dealer buys a car in Japan for 1 250 000 JPY.
The exchange rate is £1 = 145 JPY.

(a) Calculate the cost of the car in pounds (£).

£ ..... **[2]**

(b) The dealer pays shipping costs of £600.
He then sells the car in the UK for £12 500.

Calculate his percentage profit on the total cost of buying and shipping the car.

..... % **[3]**

5 Amira travels from the UK to Japan.
She changes 850 into Japanese Yen (JPY).
The bank charges a 2% fee on the amount exchanged.
The exchange rate is 1 = 180 JPY.

(a) Calculate the amount of JPY she receives.

..... JPY **[3]**

(b) While in Japan, she spends 115000 JPY.
She changes her remaining JPY back to pounds ().
The new exchange rate is 1 = 184 JPY.
The bank charges a fixed fee of 500 JPY for this transaction.
Calculate the amount of pounds she receives.

..... **[4]**

6 Carlos buys a car in the USA for \$17 400. He pays a shipping fee of \$1 200.

(a) The exchange rate is £1 = \$1.24. Calculate the total cost of the car and shipping in pounds (£).

£ ..... **[2]**

(b) Carlos sells the car in the UK for £17 100.
Calculate his percentage profit on his total cost.

..... % **[3]**

(c) Carlos changes his profit of £2 100 into euros.
The exchange rate is £1 = 1.16 euros.
The bank charges a 2% commission fee for the exchange.
Calculate the amount he receives in euros.

..... euros **[3]**

7 A trader buys 1000 items for y each.
He pays a shipping fee of 250.
He sells all 1000 items in the USA for \$4.20 each.

The exchange rate is 1 = \$1.20.
The trader makes a profit of 40% on his total cost in Euros.

Calculate the value of y.

$y =$ ..... **[5]**

8 In Year 1, the exchange rate between dollars ($) and euros (EUR) was $\$1 = x$ EUR.
In Year 2, the exchange rate changed to $\$1 = (x - 0.05)$ EUR.

In Year 1, a laptop cost 400 EUR. In Year 2, the same model laptop cost 450 EUR.

The cost of the laptop in dollars increased by $100 from Year 1 to Year 2.

(a) Write down an expression in terms of x for the cost of the laptop in dollars in Year 1.

$\$ \ldots\ldots\ldots\ldots\ldots\ldots$ **[1]**

(b) Form an equation in x and show that it simplifies to $20x^2 - 11x - 4 = 0$.

[4]

(c) Solve the equation to find the value of x.

$x = \ldots\ldots\ldots\ldots\ldots\ldots$ **[3]**

9 A shop sells laptops.
In 2022, the price of a laptop was \$840. In 2023, the price increased by 15%. In 2024, the price decreased by r% from the 2023 price. The price of the laptop in 2024 was \$821.10.

(a) Calculate the value of r.

$r =$ ..... **[3]**

(b) The shop buys 450 of these laptops from a supplier in China.
The total cost is 2 430 000 Chinese Yuan (CNY).
The exchange rate is \$1 = 7.2 CNY.

Calculate the profit the shop makes on each laptop when sold at the 2022 price of \$840.

\$ ..... **[3]**

(c) A customer in the UK buys one of these laptops in 2022 for \$840. They pay with a credit card that charges a 2.5% foreign transaction fee. The exchange rate is £1 = \$1.32.
Calculate the total amount the customer pays in pounds (£).
Give your answer correct to the nearest penny.

£ ..... **[3]**

10 **(a)** Adele buys 4 pens costing \$2.35 each and 3 notebooks costing \$4.50 each.
She pays with a \$50 note.
Work out how much change she receives.

% ..... **[3]**

(b) The exchange rate between euros () and US dollars (\$) is 1 = \$1.20.
Change 45 into US dollars.

% ..... **[1]**

11 **(a)** Tomas changes £450 into euros ().
The exchange rate is £1 = 1.18.
He spends 285 on his holiday.
He changes the remaining money back into pounds.
The new exchange rate is £1 = 1.22.
Calculate the amount of pounds he receives.
Give your answer correct to the nearest penny.

£ ..... **[4]**

(b) A television is priced at $850. In a sale, the price is reduced by 15%. A customer has a voucher that gives a further 5% discount off the sale price.
Calculate the final price paid for the television.

% ..... **[2]**

12 A shop sells two brands of coffee, Brand A and Brand B.
Brand A costs \$12.50 per 500 grams. Brand B costs \$8.40 per 300 grams.

(a) Which brand is better value for money?
You must show all your working.

..... **[3]**

(b) The shop owner mixes x grams of Brand A with y grams of Brand B to make a new blend.
She makes 2 kg of the new blend.
The total cost of the coffee in the new blend is \$54.10.

Work out the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

1 $255 \div 0.85$ **[M1]**

$\$300$ **[A1]**

$\$100$ **[A1]**

[Back to Question] *[Total Marks: 3]*

2 $300 \times 1.50 = 450$ **[M1]**

$450 - 210 = 240$ **[M1]**

$240 \div 1.50 = 160$ **[A1]**

[Back to Question] *[Total Marks: 3]*

3 (a) 850×148 **[M1]**

125800 **[A1]**

(b) $125800 - (45000 + 31500)$ or 49300 **[M1]**

$\frac{49300}{152}$ or 324.342... **[M1]**

324.34 **[A1]**

[Back to Question] *[Total Marks: 5]*

4 (a) $1\,250\,000 \div 145$ **[M1]**

8620.69 **[A1]**

(b) Total cost $= 8620.69 + 600 = 9220.69$ **[M1]**

Profit $= 12\,500 - 9220.69 = 3279.31$ (or $\frac{12500 - 9220.69}{9220.69}$) **[M1]**

35.6 (accept $35.56\ldots$) **[A1]**

[Back to Question] *[Total Marks: 5]*

5 (a) 850×0.98 or $850 - (850 \times 0.02)$ **[M1]**

833 **[A1]**

149940 **[A1]**

(b) $149940 - 115000$ $(= 34940)$ **[M1]**

their $34940 - 500$ $(= 34440)$ **[M1]**

their $34440 \div 184$ **[M1]**

187.17 (accept 187.17...) **[A1]**

[Back to Question] *[Total Marks: 7]*

6 (a) $\frac{17400 + 1200}{1.24}$ or $\frac{18600}{1.24}$ **[M1]**

15000 **[A1]**

(b) $17100 - 15000$ or 2100 **[M1]**

$\frac{2100}{15000} \times 100$ **[M1]**

14 **[A1]**

(c) $2100 \times (1 - 0.02)$ or 2100×0.98 or 2058 **[M1]**

2058×1.16 **[M1]**

2387.28 **[A1]**

[Back to Question] *[Total Marks: 8]*

7 Total revenue in USD $= 1000 \times 4.20 = 4200$ **[B1]**

Convert revenue to EUR: $4200 \div 1.20$ $(= 3500)$ **[M1]**

Equate to cost with profit multiplier: Total Cost $\times 1.40 = 3500$ or $3500 \div 1.40$ $(= 2500)$ **[M1]**

Form equation for cost: $1000y + 250 = 2500$ **[M1]**

$y = 2.25$ **[A1]**

[Back to Question] *[Total Marks: 5]*

8 (a) $\frac{400}{x}$ **[B1]**

(b) $\frac{450}{x - 0.05} - \frac{400}{x} = 100$ **[M1]**

$450x - 400(x - 0.05) = 100x(x - 0.05)$ **[M1]**

$450x - 400x + 20 = 100x^2 - 5x$ **[M1]**

$50x + 20 = 100x^2 - 5x \implies 100x^2 - 55x - 20 = 0$ **[A1]**

Dividing by 5 yields $20x^2 - 11x - 4 = 0$ (must show working to reach final equation) **[A1]**

(c) $(5x - 4)(4x + 1) = 0$ or use of quadratic formula $\frac{11 \pm \sqrt{(-11)^2 - 4(20)(-4)}}{2(20)}$ **[M2]**

If M2 not scored, allow [M1] for $(5x \pm a)(4x \pm b) = 0$ where $ab = 4$ or $a(4) \pm b(5) = -11$, or correct substitution into formula but with one error.

0.8 (with -0.25 rejected or ignored) **[A1]**

[Back to Question] *[Total Marks: 8]*

9 (a) 840×1.15 or 966 **[M1]**

$\frac{821.10}{966}$ or 0.85 **[M1]**

15 **[A1]**

(b) $\frac{2430000}{450}$ or 5400 **[M1]**

$\frac{5400}{7.2}$ or 750 **[M1]**

90 **[A1]**

(c) 840×1.025 or 861 **[M1]**

$\frac{861}{1.32}$ or 652.272... **[M1]**

652.27 **[A1]**

[Back to Question] *[Total Marks: 9]*

10 **(a)** Cost of pens $= 4 \times 2.35 = 9.40$ **[M1]**

Cost of notebooks $= 3 \times 4.50 = 13.50$ **[M1]**

Change $= 50 - (9.40 + 13.50) = 27.10$ **[A1]**

(b) 54 **[B1]**

[Back to Question] *[Total Marks: 4]*

11 **(a)** Amount in euros $= 450 \times 1.18 = 531$ **[M1]**

Remaining euros $= 531 - 285 = 246$ **[M1]**

Amount back in pounds $= 246 \div 1.22$ **[M1]**

201.639... rounding to 201.64 **[A1]**

(b) Sale price $= 850 \times 0.85$ $(= 722.5)$ **[M1]**

Final price $= 722.5 \times 0.95 = 686.375$ which rounds to 686.38 **[A1]**

[Back to Question] *[Total Marks: 6]*

12 **(a)** Cost of Brand A per 100g $= 12.50 \div 5 = 2.50$ (or per gram $= 0.025$) **[M1]**

Cost of Brand B per 100g $= 8.40 \div 3 = 2.80$ (or per gram $= 0.028$) **[M1]**

Brand A with correct supporting working **[A1]**

(b) $x + y = 2000$ **[M1]**

Cost equation: $0.025x + 0.028y = 54.10$ (or equivalent, e.g. $\frac{12.5}{500}x + \frac{8.4}{300}y = 54.1$) **[M1]**

Substitute $y = 2000 - x$: $0.025x + 0.028(2000 - x) = 54.10$ **[M1]**

$0.025x + 56 - 0.028x = 54.10 \implies -0.003x = -1.9 \implies x = 633.3...$ and $y = 1366.6...$

Accept $x = 633$ and $y = 1367$ (or exact fractions $\frac{1900}{3}$ and $\frac{4100}{3}$) **[A1]**

[Back to Question] *[Total Marks: 7]*